Mehrdad Mirzaei

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Summary

AI/ML Engineer with a Ph.D. in Computer Science and extensive experience designing, building, and deploying production-grade AI and Generative AI solutions. Specialized in Retrieval-Augmented Generation (RAG), LLM-powered applications, agentic workflows, LLM fine-tuning, and scalable machine learning pipelines across AWS and Azure. Experienced in leading end-to-end development of AI solutions for natural language interfaces, intelligent automation, and decision support, from architecture and model development through deployment and productionization.

Professional Experience

Generative AI Engineer, *CGI Inc.* *(Client: Fannie Mae) - Reston, VA* *Feb 2025 – Present*

- Developed an AI control-testing platform using FastAPI and AWS Strands Agents that orchestrates specialist LLM agents through phase-gated planning and testing workflows to automate control testing.
- Built an evaluation framework for an AI control-testing service, validating contract compliance and response quality with deterministic test suites and production-grade judge model assessments.
- Developed a Text-to-SQL chatbot on AWS SageMaker using Streamlit and AWS Bedrock, leveraging Retrieval-Augmented Generation (RAG) with Bedrock Knowledge Base to convert prompts into SQL.
- Designed and implemented an agentic workflow using LangGraph to automate data extraction from PDF files, using LLMs to streamline and validate the loan application certification process.
- Fine-tuned large language models using LoRA on Azure Machine Learning (Azure ML) with question–answer datasets, improving the accuracy of the question-answering application by 20%.

Machine Learning Engineer, *ReadyOn AI* - *San Francisco, CA* *Jan 2024 – Jan 2025*

- Developed a scalable machine learning pipeline on Azure ML for predictive modeling, supporting workforce optimization and operational decision-making.
- Built data processing pipelines using Azure Databricks (Spark) to handle large-scale time-series data for feature engineering and model training.
- Designed and evaluated models using statistical analysis and experimentation (A/B testing, causal inference) to validate model performance and business impact.

Machine Learning Engineer, *Epsilon* - *Chicago, IL* *April 2021 – Sept 2023*

- Developed and deployed machine learning models (Logistic Regression, XGBoost) for customer conversion prediction, improving targeting strategies and marketing effectiveness.
- Designed and implemented an end-to-end customer segmentation pipeline using clustering algorithms (K-means), enabling the identification of customer groups and supporting targeted marketing strategies.
- Applied autoencoder models in PyTorch for feature learning and dimensionality reduction on high-dimensional structured data, improving feature representation and predictive modeling tasks.

Internship

Machine Learning Intern, ***AT&T*** - *Middletown, NJ* *Oct 2016 – Oct 2018*

- Implemented a machine learning pipeline for failure prediction using PySpark, helping reduce unnecessary physical dispatches for AT&T home services.
- Designed deep learning models using RNN and CNN architectures to identify faulty residential gateways, achieving high prediction accuracy.
- Applied Natural Language Processing (NLP) techniques to analyze large volumes of customer feedback and communication data, extracting insights to support improvements in customer service strategies and product offerings.

Software Engineer Intern, ***The ILS Institute at SUNY*** - *Albany, NY* *Feb 2014 – May 2016*

- Designed and implemented scalable text processing pipelines in Java for large-scale data analysis, with a focus on performance and maintainability.
- Developed data pipelines using Java and SQL to process and analyze gameplay data, improving data accessibility and system efficiency.
- Built efficient, scalable algorithms in Java to process multi-party conversation data, optimizing memory usage and query performance in high-volume environments.

Skills

- **Programming Languages**: Python, SQL, Java, Scala
- **Generative AI**: Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings, Vector Databases, LLM Fine-Tuning, LLM Evaluation, Agentic Workflows
- **AI Frameworks & Tools**: LangChain, LangGraph, AWS Strands Agents, MCP, FastAPI, Streamlit
- **Large Language Models**: GPT, Claude, Gemini, Llama
- **Machine Learning**: PyTorch, TensorFlow, Scikit-learn, XGBoost, Logistic Regression, Random Forest, K-means, Transformer Models
- **Cloud & AI Platforms**: AWS (Bedrock, SageMaker, Textract), Azure (Azure ML, Azure OpenAI, Data Factory), GCP
- **Data Engineering & Processing**: Spark, Databricks, Airflow, Pandas, NumPy
- **MLOps & Infrastructure**: Docker, Kubernetes, MLflow, CI/CD, GitLab, GitHub, Terraform
- **Databases & Data Warehouses**: PostgreSQL, MySQL, MongoDB, DynamoDB, Redshift

Education

- SUNY at Albany, *Albany, NY* - ***Ph.D. Computer Science*** *Sept 2020*
- University of Tehran, *Tehran* - ***M.Sc. Software Engineering*** *Sept 2012*
- Shahid Beheshti University, *Tehran* - ***B.Sc. Software Engineering*** *Sept 2009*